Microscopic Traffic Flow Modelling (Part I: Introduction & Basics)

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Semester 2 2025

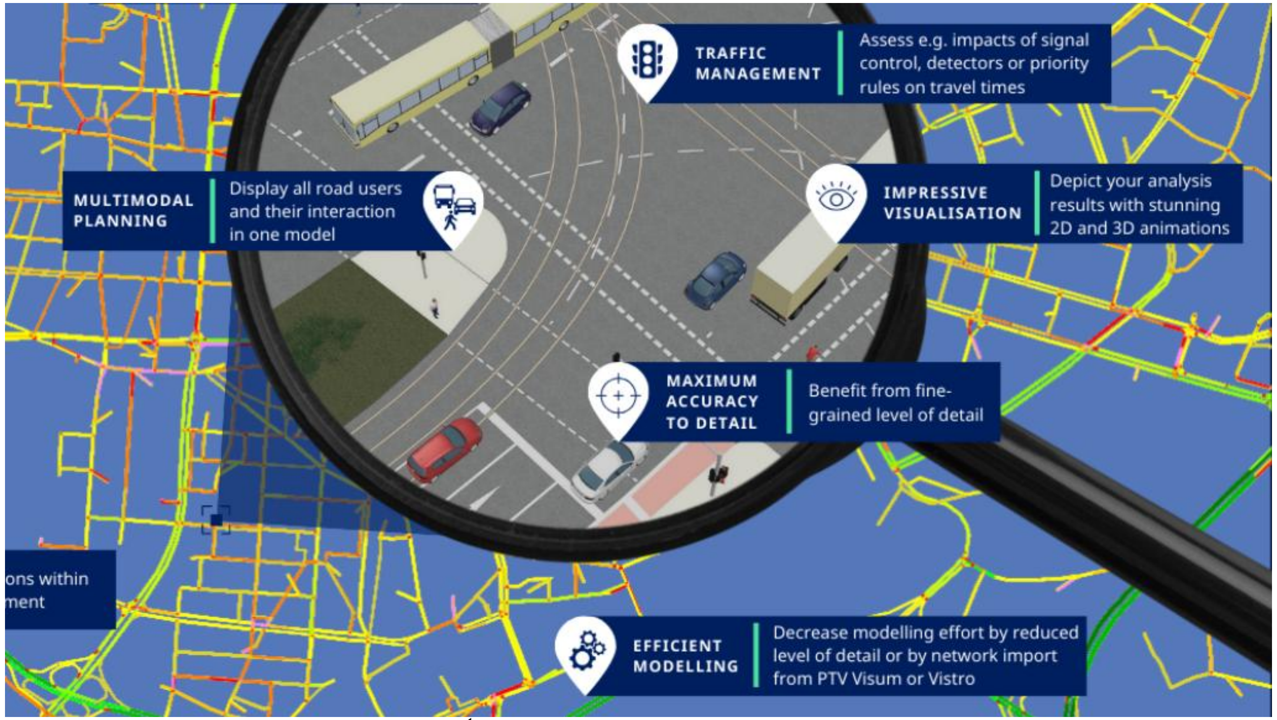
Agenda

- Overview of Microscopic Traffic Flow Modelling
- Car Following Modelling Basics

Overview of Microscopic Traffic Flow Modelling

Levels of Traffic Modelling

- Microscopic (Vehicles & Drivers)
  - ✓ Models each vehicle/driver behavior (car-following, lane-changing).
  - ✓ Detailed simulation, intelligent transport systems, local junction analysis.
- Macroscopic (Traffic Flows)
  - ✓ Treats traffic like fluid flow (density, speed, flow).
  - ✓ Network modelling, strategic planning, regional/national models.
- Mesoscopic (Vehicle Groups)
  - ✓ Models flows of vehicles in groups with some individual characteristics.
  - ✓ Corridor-level studies, intermediate detail.



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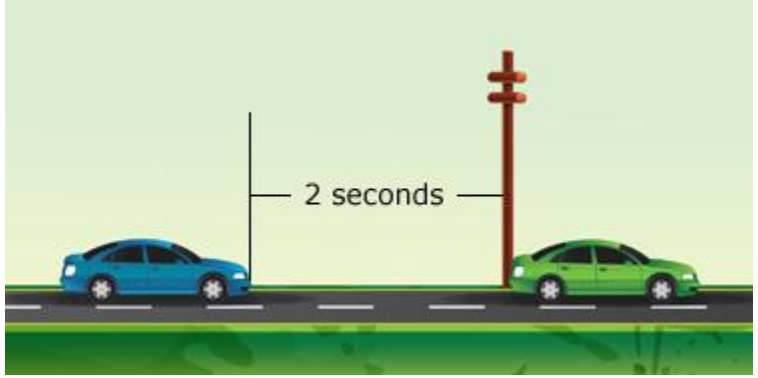
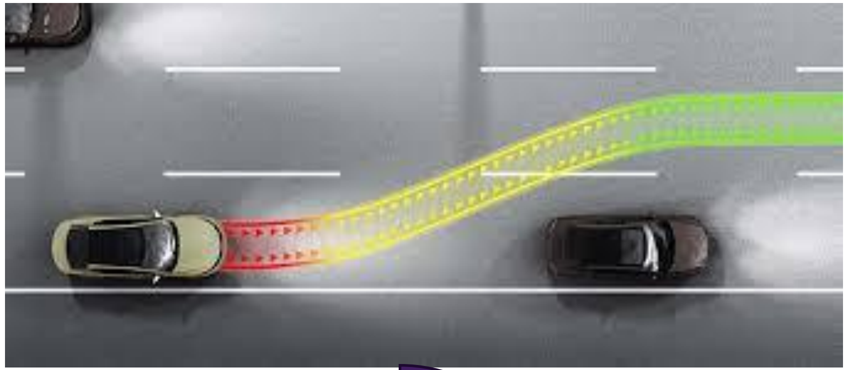
Characterisation of Microscopic Models

- Microscopic models considers the smallest objects that play an important role in a system, e.g., atoms in physics, individual decision makers in economics.
- In traffic flow, this smallest object usually is **the driver-vehicle unit** but it can also be a cyclist, a pedestrian, etc.
- Microscopic models are more detailed than the macroscopic ones which locally aggregate the microscopic quantities.
- Microscopic models are less detailed than models for the vehicle dynamics which include details related to brake, engine control path, stability control, etc.

Two primary tasks of a driver-vehicle unit





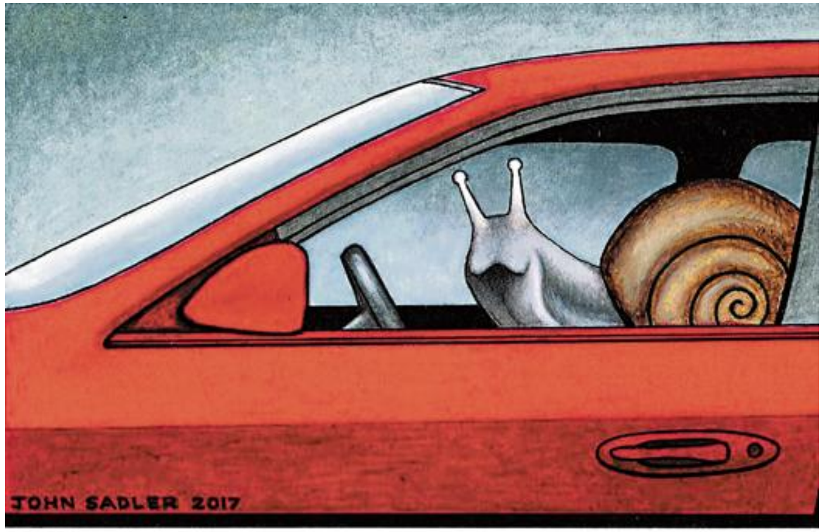
Car following

Lane changing

Every rational driver can be a good car-following modeller!

A good starting point is to think how you normally drive.

Safety: Not too close



Efficiency: Not too far away

What makes car following modelling complicated is due to the simple fact

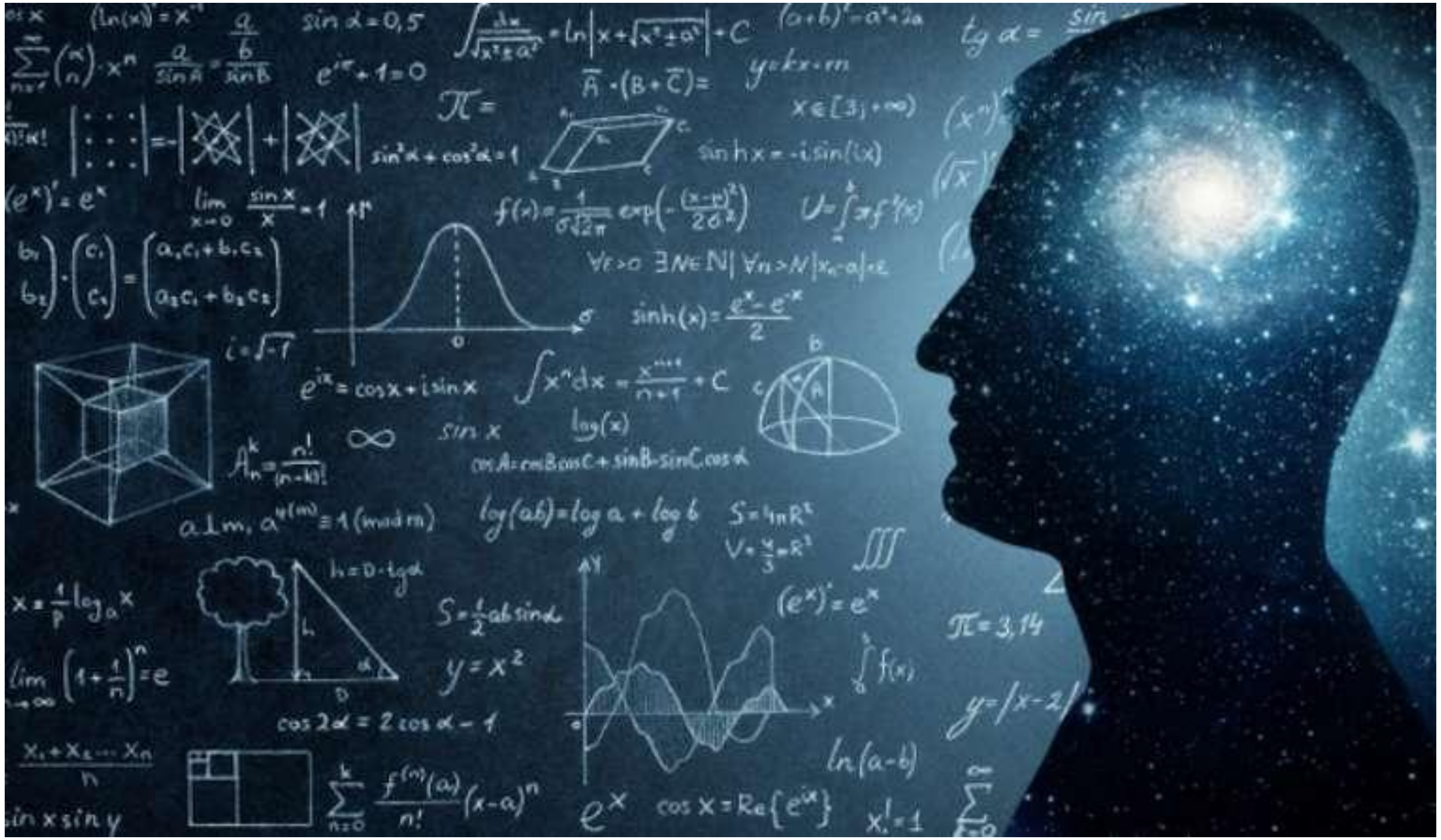
Each driver is different!

Car following in real life

Source: online

A traffic congestion that seemingly comes from nowhere

So, we need properly developed car-following models



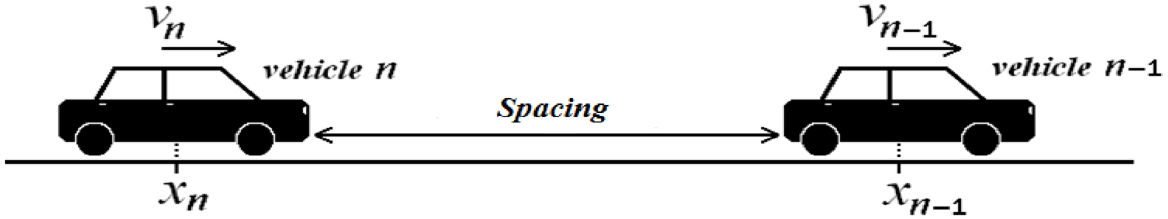
Car Following Modelling Basics

Introduction

- A large number of CF models have been developed in an attempt to describe CF behavior under a wide range of traffic conditions, ranging from free-flow to extreme situations.
- In a strict sense, car-following models describe the driver's behavior only in the presence of interactions with other vehicles while free traffic flow is described by a separate model. In a more general sense, car-following models include all traffic situations such as car-following situations, free traffic, and also stationary traffic.
- A car-following model is ***complete*** if it is able to describe all situations including acceleration and cruising in free traffic, following other vehicles in stationary and non-stationary situations, and approaching slow or standing vehicles, and red traffic lights.

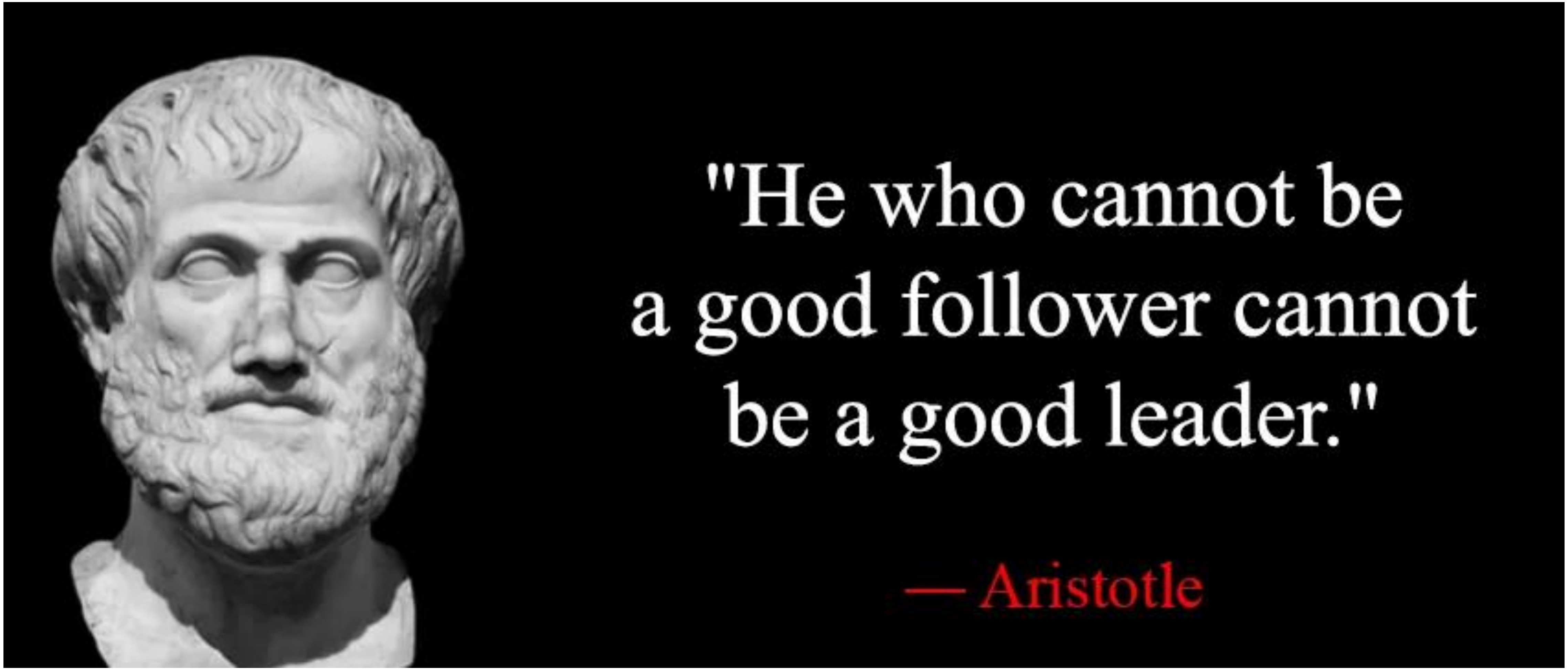
Car-following model

- **Car-following (CF) model** describes the decision of the driver to follow the preceding vehicle **efficiently** and **safely**.

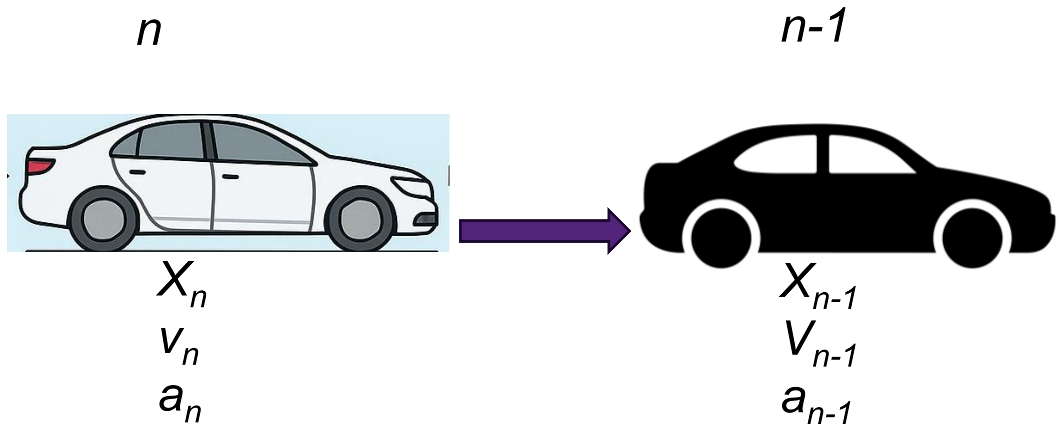


- Rules governing CF are typically simplified as a set of mathematical models that aim for reproducing the longitudinal interactions in the driver-vehicle system.
- As one of the oldest topics in traffic engineering, numerous CF models have been proposed in the literature.

Think about how you follow a car in front of you



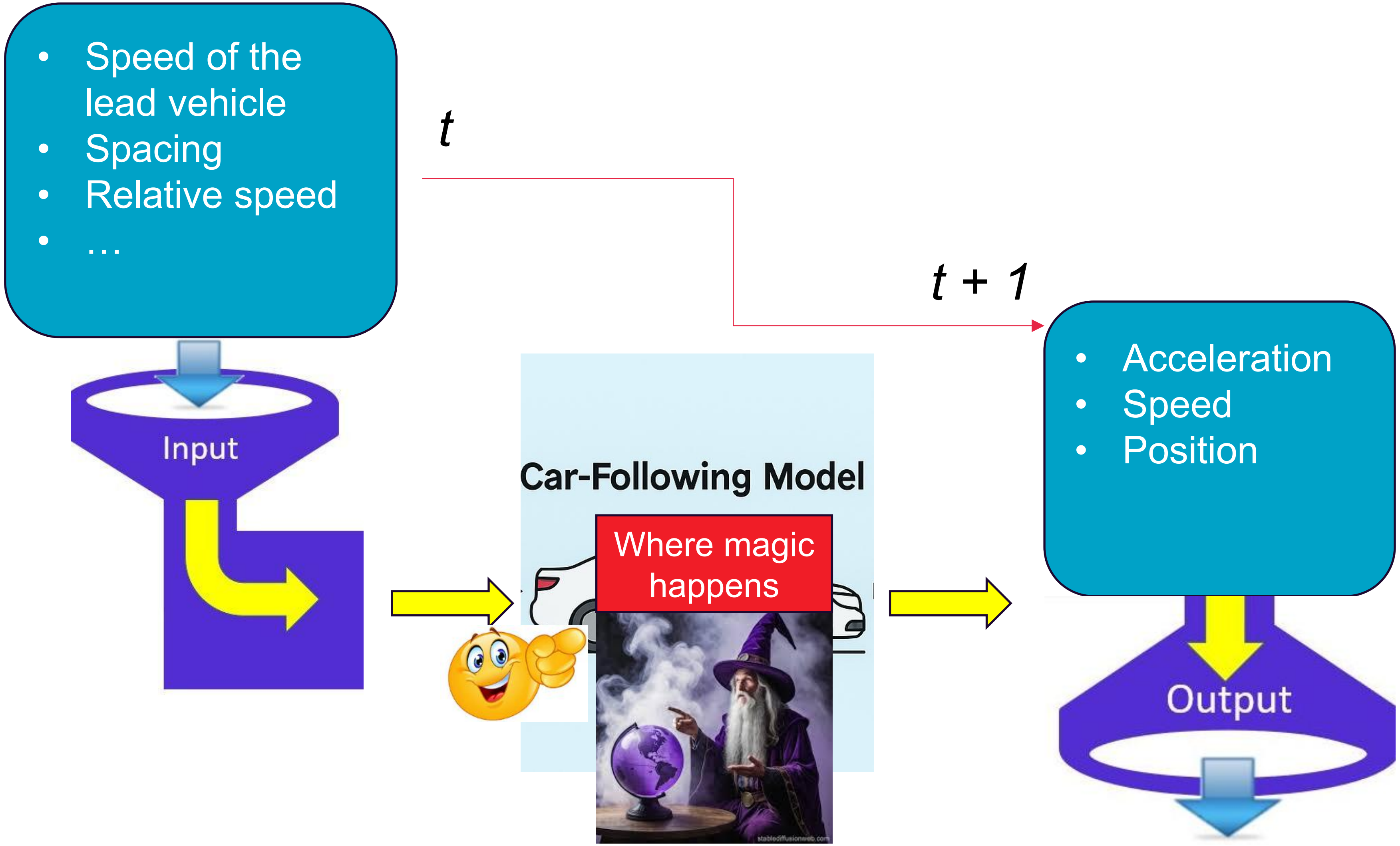
The state variables of car-following models



Each driver-vehicle unit n is described by the state variables:

- location $x_n(t)$, and speed $v_n(t)$ as a function of the time t, and their derived variables such as spacing and relative speed.
- by the attribute "vehicle length" l_n .

We define the vehicle index n such that vehicles pass a stationary observer (or detector) in ascending order, i.e., the first vehicle has the lowest index.

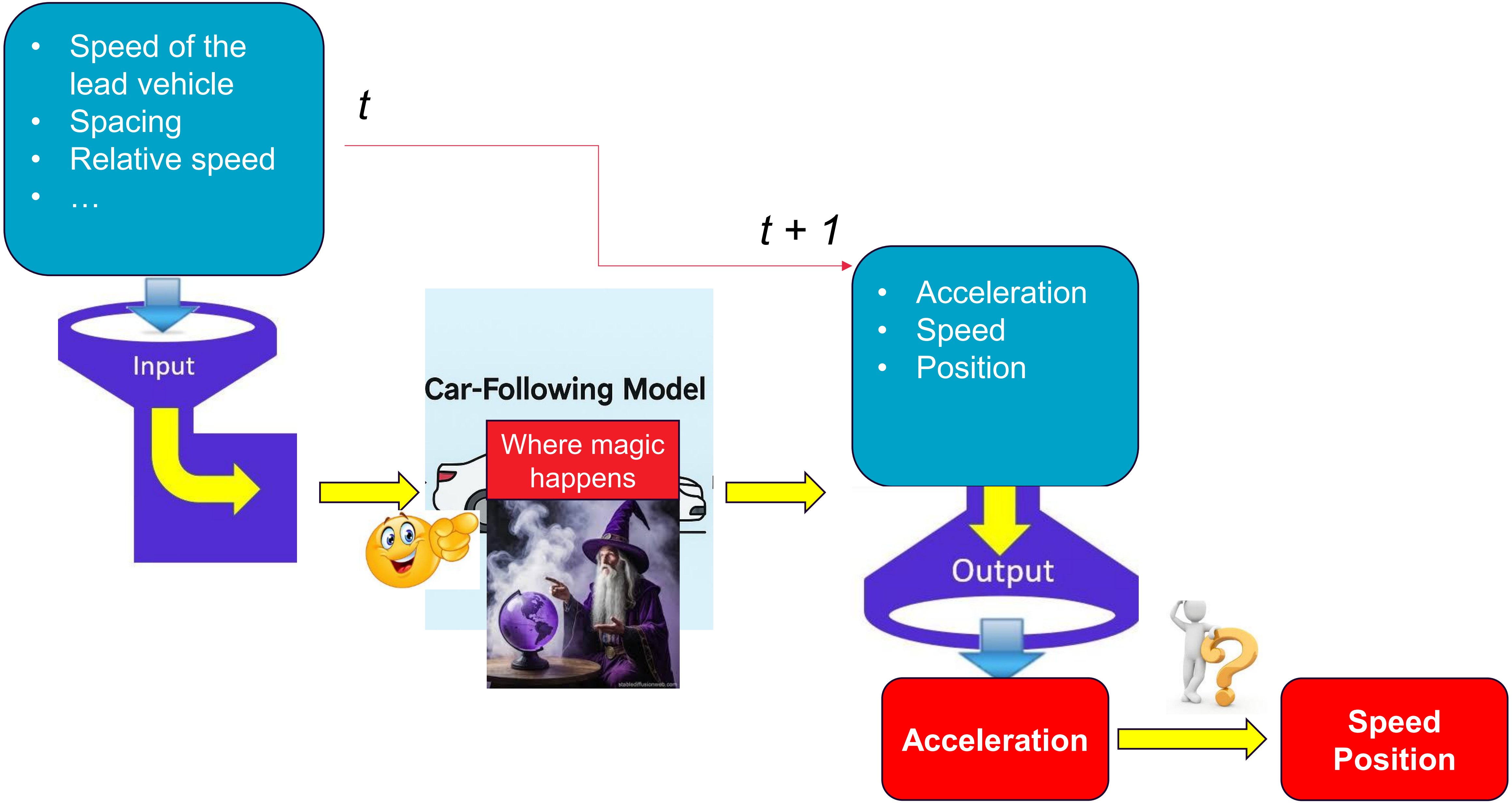

Speed of the lead vehicle
Spacing
Relative speed
…
Input
t
t + 1
Car-Following Model
Where magic happens
Acceleration
Speed
Position
Output

Continuous-time car-following models

In continuous-time models, the driver's response is directly given in terms of an acceleration function $a_{mic}(s_n, v_n, v_{n-1})$, leading to a set of coupled ordinary differential equations:

$$\dot{x}_n(t) = \frac{dx_n(t)}{dt} = v_n(t)$$

$$\dot{v}_n(t) = \frac{dv_n(t)}{dt} = a_{mic}(s_n, v_n, v_{n-1}) = a_{mic}(s_n, v_n, \Delta v_n)$$



Speed of the lead vehicle
Spacing
Relative speed
…
t
t + 1
Input
Car-Following Model
Where magic happens
Acceleration
Speed
Position
Output
Acceleration
Speed
Position

Numerical integration

- In general, time-continuous models cannot be solved analytically and an integration scheme is necessary for an approximate numerical solution of the system of equations.
- For traffic flow applications, only **explicit update schemes** with a fixed time step are practical.
- Assuming a constant update time step Δt, a simple but efficient explicit update method is given by the "ballistic" assumption of **constant accelerations during each time step**.

$$v_n(t+\Delta t) = v_n(t) + a_{mic}(s_n(t), v_n(t), \Delta v_n(t))\Delta t$$

$$x_n(t+\Delta t) = x_n(t) + \frac{v_n(t) + v_n(t+\Delta t)}{2}\Delta t$$

Discrete-time CF models

- Time is not modelled as a continuous variable but discretized into finite and generally constant time steps.
- The driver's response is no longer modelled by an acceleration function but by **a speed function**, indicating the speed that will be reached at the end of the next time step.
- A generic formulation of discrete-time CF models is shown:

$$v_n(t + \Delta t) = v_{mic}(s_n(t), v_n(t), v_{n-1}(t))$$

Discrete-time CF models

- Compared to continuous models, discrete-time car-following models are generally less realistic and less flexible but require less computing power for their numerical integration.
- Most discrete-time car-following models have been proposed at times where computing was more expensive. Nowadays, hundreds of thousand vehicles can be simulated with time-continuous models on a PC in real time, so this numerical advantage become less relevant.
- Most traffic simulation software uses time-continuous models.

Different CF models

- A specific time-continuous model is uniquely characterized by **its acceleration function a_{mic}**.
- Similarly, a specific discrete-time model is completely characterized by **its speed function v_{mic}**.
- When simulating heterogeneous traffic consisting of a variety of driving styles and vehicle classes (such as cars and trucks), each driver-vehicle combination is described by different acceleration functions $a_{mic}^{n}(s, v, v_l)$ or speed function $v_{mic}^{n}(s, v, v_l)$, respectively.

Prove the equivalence between continuous-time models and discrete-time models if the ballistic update scheme is used in updating continuous-time models.

Mathematical equivalence between continuous-time and discrete-time CF

If we assume $a_{mic}(s, v, v_l) = {}^{(v_{mic}(s,v,v_l)-v)}/_{\Delta t}$,

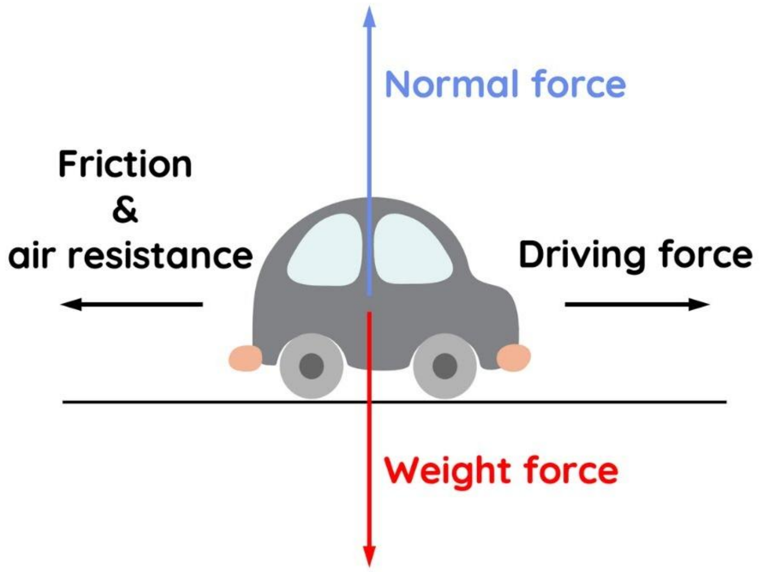
The combination of a continuous-time model with the ballistic update scheme is **mathematically equivalent** to discrete-time models.

Conceptual difference between continuous-time and discrete-time CF

- For discrete-time models, the time step Δt plays the role of a model parameter typically describing the reaction time, the time headway, or the speed adaptation time.
- For time-continuous models, the update time Δt is an auxiliary variable of the approximate numerical solution which preferably should be small as the true solution is obtained in the limit $\Delta t \to 0$ (at least if the numerical method is consistent and stable).

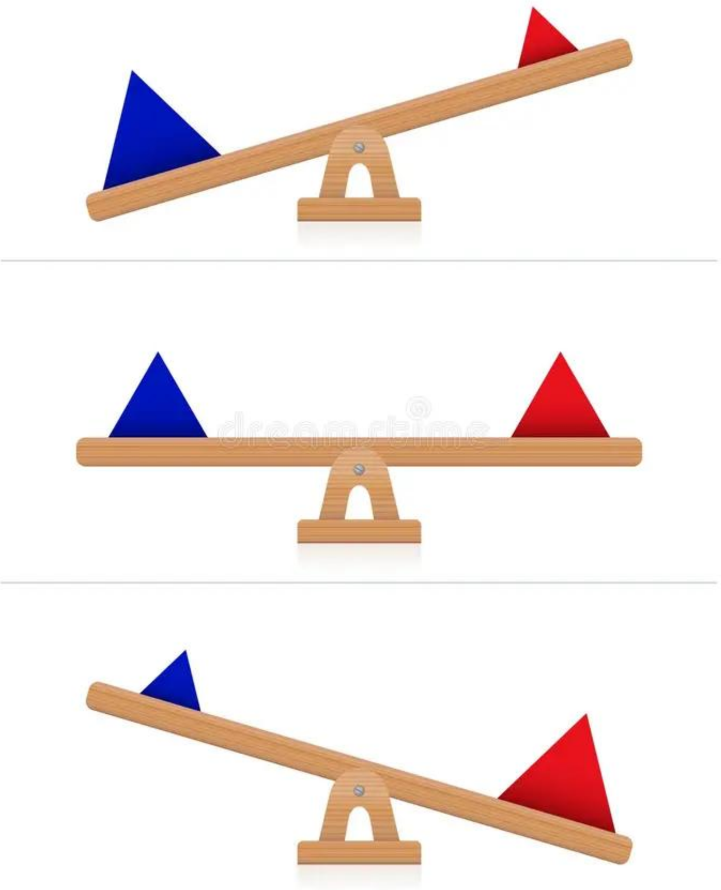
Steady State Equilibrium

- Since the driver-vehicle units of microscopic models are equivalent to *driven particles* of physical systems, there is no equilibrium in the strict sense. Instead, there is a stationary state where the forces and the entering and exiting energy fluxes are balanced.
- Strictly physically, this can be interpreted in terms of a balance of the forces (the sum of friction, wind drag, and engine driving force equates to zero) or energy fluxes (engine power equals change in potential and kinetic energy plus energy dissipation rate by friction and wind drag).



Steady State Equilibrium: balancing the social forces

- More relevant for traffic flow, however, is the concept of balancing the *social forces*:
  - ✓ The desire to go ahead generates a positive (accelerating) social force while the interactions with other vehicles generally lead to negative social forces in order to avoid critical situations and crashes.
- In any case, such a balanced state is denoted as ***steady-state equilibrium***. For microscopic models, a consistent description of the steady-state equilibrium requires **identical** driver-vehicle units on a **homogeneous road**.



Conditions of Steady-State Equilibrium

- Technically, this implies that the model parameters are the same for all drivers and vehicles, i.e., the acceleration or speed functions characterizing the respective model do not depend on the vehicle index, $a^i_{mic}(s, v, v_l) = a_{mic}(s, v, v_l)$, and $v^i_{mic}(s, v, v_l) = v_{mic}(s, v, v_l)$, respectively.
- From the modelling point of view, the steady-state equilibrium is characterised by the following two conditions:
  - ✓ Homogeneous traffic: All vehicles drive at the same speed ($v_i = v$) and keep the same gap behind their respective leaders ($s_i = s$).
  - ✓ No accelerations: $\dot{v}_i = 0$ or $v_i(t + \Delta t) = v_i(t)$ for all vehicles i.

Derive the Steady-state Relations of a CF Model

- For time-continuous models, this implies $a_{mic}(s, v, v_l) = 0$,
- For discrete-time models, this implies $v_{mic}(s, v, v_l) = v$
- Depending on the model, the microscopic steady-state relations above can be solved for
  - ✓ The equilibrium speed $v_e(s)$ as a function of the gap;
  - ✓ The equilibrium gap $s_e(v)$ for a given speed.

Bridging Microscopic and Macroscopic Relations

- The key is to use the relation between the distance gap *s* and the density *k*: $s_i = s = \frac{1}{k} - l$
- Meanwhile, the steady-state equilibrium implies that the speed of all vehicles is the same and equal to the macroscopic speed $V(x,t) = v_i(x,t) = v_e(s)$.
- So, we can derive the steady-state speed-density relation and the fundamental diagram:

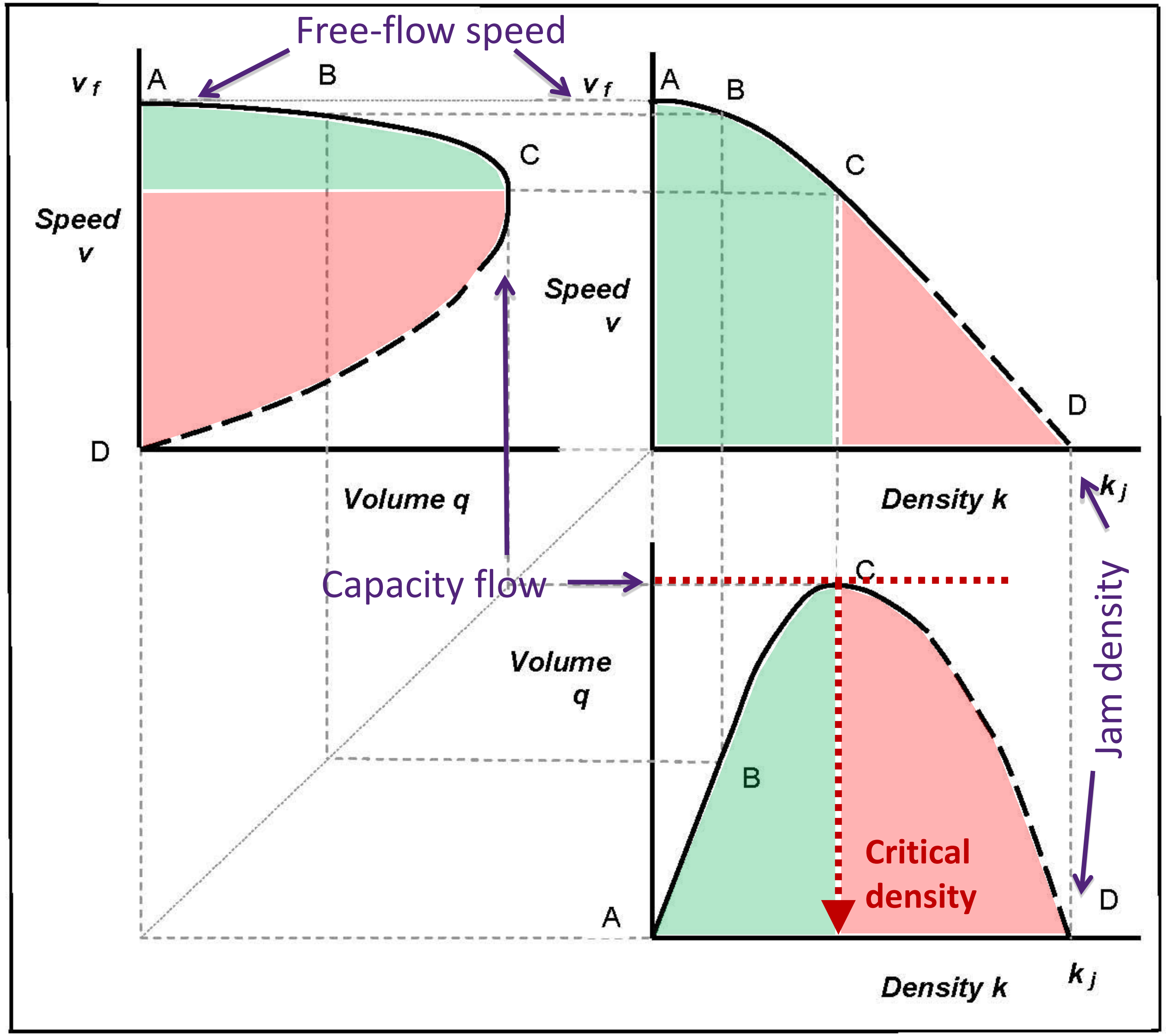
$$V_e(k) = v_e\left(\frac{1}{k} - l\right),\ Q_e(k) = k v_e\left(\frac{1}{k} - l\right).$$

Flow-density-speed relationships

Fundamental Diagrams (FD)

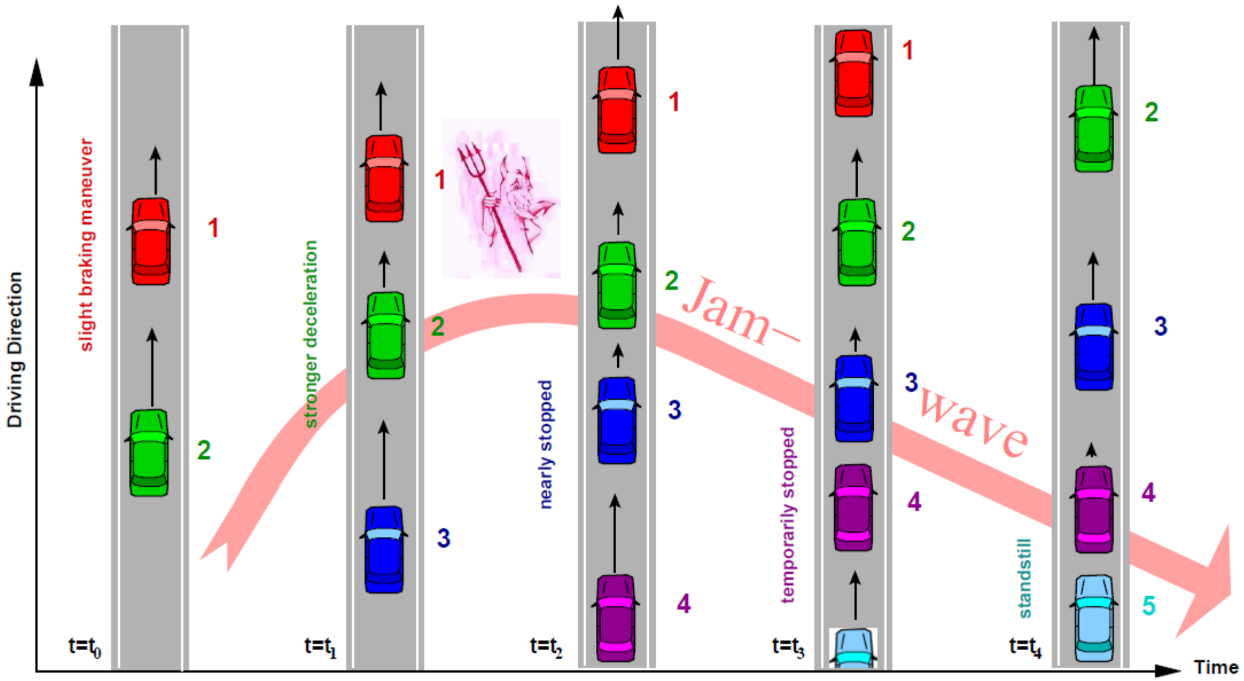
‘Fundamental relationships’ of ‘uninterrupted’ traffic flow

At **A**, no interaction
A-**B**, free flow
B-**C**, steady flow
At **C**, traffic unstable
C-**D**, congested flow
At **D**, traffic stationary



Discussion: how to simulate heterogeneous traffic using car following model?

Stability Analysis



The vicious circle: In order to regain the safety gap, the driver of every following vehicle needs to brake harder than his or her predecessor. The numbers beside the vehicles denote the vehicle index.

A traffic congestion that seemingly comes from nowhere

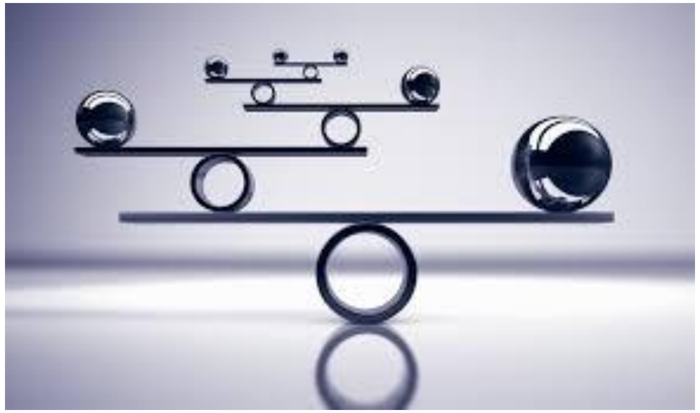
Phantom Traffic Jam Explained.

The Japanese Experiment.

Phantom Intersection.

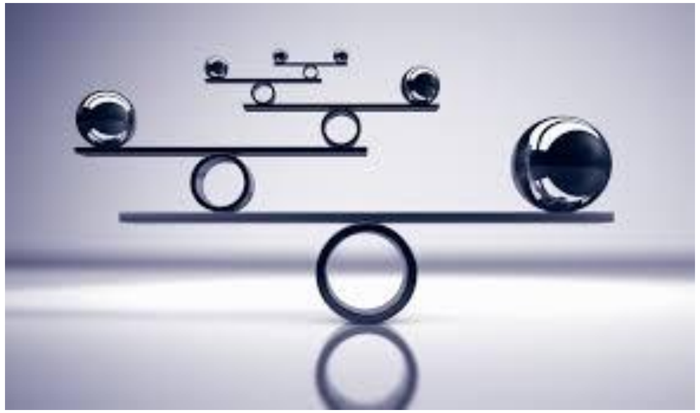
Local stability

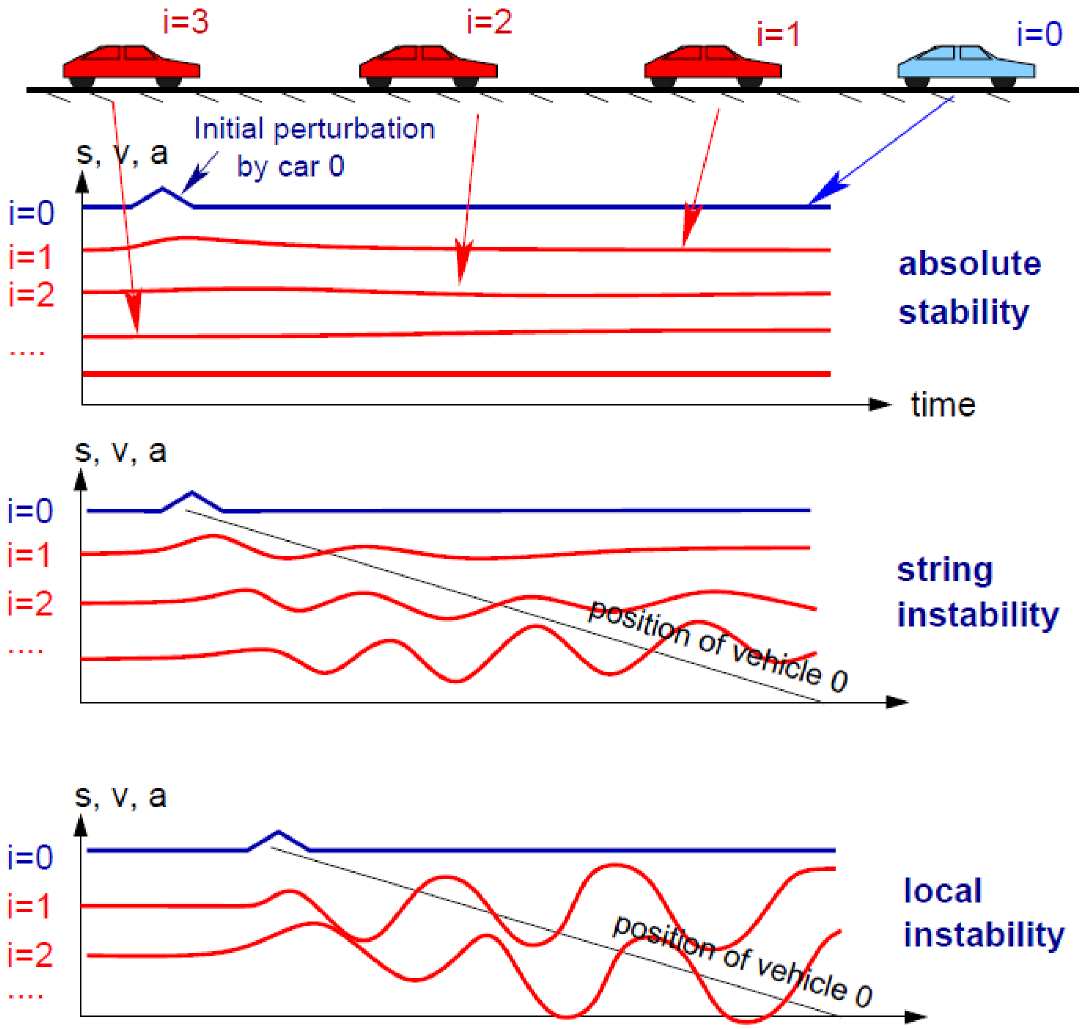
- Local stability is to investigate the stability of a single vehicle's movement over time under the influence of a small perturbation that is often originated from the leading vehicle's movement.
- It is locally stable if the perturbation diminishes over time, which corresponds to asymptotic stability in dynamic systems.



String stability

- String stability focuses on the stability of a platoon of vehicles over space under the influence of a small perturbation that is originated from the first vehicle of the platoon.
- It is asymptotically stable if the perturbation strictly diminishes over space/vehicles.





i=3
i=2
i=1
i=0
Initial perturbation
by car 0
s, v, a
i=0
i=1
i=2
....
absolute
stability
time
s, v, a
i=0
i=1
i=2
....
position of vehicle 0
string
instability
s, v, a
i=0
i=1
i=2
....
position of vehicle 0
local
instability

For a good CF model,
1) shall it be locally stable?
2) shall it be string unstable?

Implications on CF Modelling

- Being locally stable is an important feature that a CF model should have because typically a driver is able to easily recover from small disturbances in real traffic and return to a steady CF state over time.
- The string stability analysis is particularly relevant to traffic flow modelling as the stop-and-go oscillation in real traffic is a manifestation of the string instability of traffic flow.
- From the CF model development perspective, for a realistic CF model, it should be able to depict the string instability in order to replicate characteristics related to traffic oscillations;
- However, from traffic operation perspective, string instability should be minimized or avoided. Thus, the string stability analysis has been frequently implemented since the early-stage of CF model development.

Further readings

Chapter 10, Traffic flow dynamics: Data, models and simulation by Martin Treiber and Arne Kesting.

van Wageningen-Kessels, F., van Lint, H., Vuik, K. et al., 2015. *EURO J Transp Logist* 4: 445. https://doi.org/10.1007/s13676-014-0045-5

Saifuzzaman, M., Zheng, Z., 2014. Incorporating human-factors in car-following models: A review of recent developments and research needs. *Transportation Research Part C* 48, 379 - 403.

Thank you

Questions?